Roll No. ....

TME–305

B. TECH. (ME) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

MANUFACTURING PROCESSES—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) A certain mold has a sprue whose length is 20 cm and the cross-sectional area at the base of the sprue is 2.5 cm^2. The sprue feeds a horizontal runner leading into a mold cavity whose volume is 1560 cm^3. Determine : (i) velocity of the molten metal at the base of the sprue, (ii) volume rate of flow, and (iii) time to fill the mold. (CO1)

OR

(b) A cylindrical-shaped part is to be cast out of aluminium. The radius of the cylinder r = 250 mm and its thickness h = 20 mm. If the mold constant C_m = 2.0 sec/mm^2 in Chvorinov's rule, how long will it take the casting to solidify ? (CO1)

2. (a) A hollow casting is produced using a cylindrical core of H = D = 100 mm. Density of liquid metal is 2700 kg/m^3 and density of core metal is 1600 kg/m^3. What is the net buoyancy force on the core ? (CO1)

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OR

(b) Determine the permeability of moulding sand when it take 1 min. 25 seconds to allow 2000 cm^3 of air through a standard cylindrical specimen of 5.08 cm height at a pressure difference of 5 gm/cm^3. (CO1)

3. (a) Discuss the shrinkage, machining and draft allowance in brief. (CO1)

OR

(b) Write a short note on types of patterns in metal casting. (CO1)

4. (a) Discuss in brief the gating element used in metal casting. (CO1)

OR

(b) With the help of a diagram, explain investment casting. (CO1)

5. (a) Write a short note on casting defects. (CO1)

OR

(b) Explain the mechanism of drop forging. (CO2)